

Homework 8

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The link to the Github repo containing the R file can be found [here](#).

Problem 1

Part A

Table 1: Coefficients for Regression: Creatinine Clearance vs. Age

Term	Estimate
(Intercept)	147.813
age	-0.620

By using a linear regression model predicting creatine levels based off of age, the y-intercept is 147.8 and the slope is -.62. Plugging in these values for a 55 year old, we should expect a creatine level of $(147.8 - 55 * (.62)) = 113.7 \text{ mL/min}$.

Part B

Creatine clearance decreases as people get older by a rate of .62 mL/min per year. This is determined by observing the slope of the regression graph.

Part C

Using the intercept of 147.8 and slope of -.62., we can determine the expected creatine level for each age. A 40 year old has an expected creatine level of $(147.8 - 40(.62)) = 123.0 \text{ mL/min}$, and a 60 year old has an expected creatine level of $(147.8 - 60 * (.62)) = 110.6 \text{ mL/min}$.

Once we have these values, we can subtract the actual values to find which is higher. A 60 year old with a creatine level of 112 mL/min is $112 - 110.6 = 1.4 \text{ mL/min}$ higher than expected, but a 40 year old with a creatine level of 135 mL/min is $135 - 123 = 12 \text{ mL/min}$ higher than expected. Since $12 > 1.4$ it is concluded that the 40 year has a higher creatine level for his age and is thus healthier.

Problem 2

Part 1

To find the growth rate and doubling times of each country, I first separated the dataset into each country, then applied an exponential growth model to each country's death counts. After producing the model, each dataset was resampled 10,000 times in a Monte Carlo simulation, with the doubling time and growth rate being recorded for each iteration. This allowed me to form a confidence interval for each value.

To find the doubling time, I used the equation

$$\text{Doubling Time} = \frac{\log(2)}{r}$$

The reason I didn't use 70 as the numerator is because r is already a logarithmic rate, rather than a percentage of growth.

These are the values for Italy:

Italy's Growth Rate: 0.183

Italy's Growth Rate 95% Confidence Interval: [0.16, 0.208]

Italy's Doubling time: 3.8 days

Italy's Doubling time 95% Confidence Interval: [3.3, 4.3]

Part 2

Like with Italy, the Spain estimates are created using a Monte Carlo simulation.

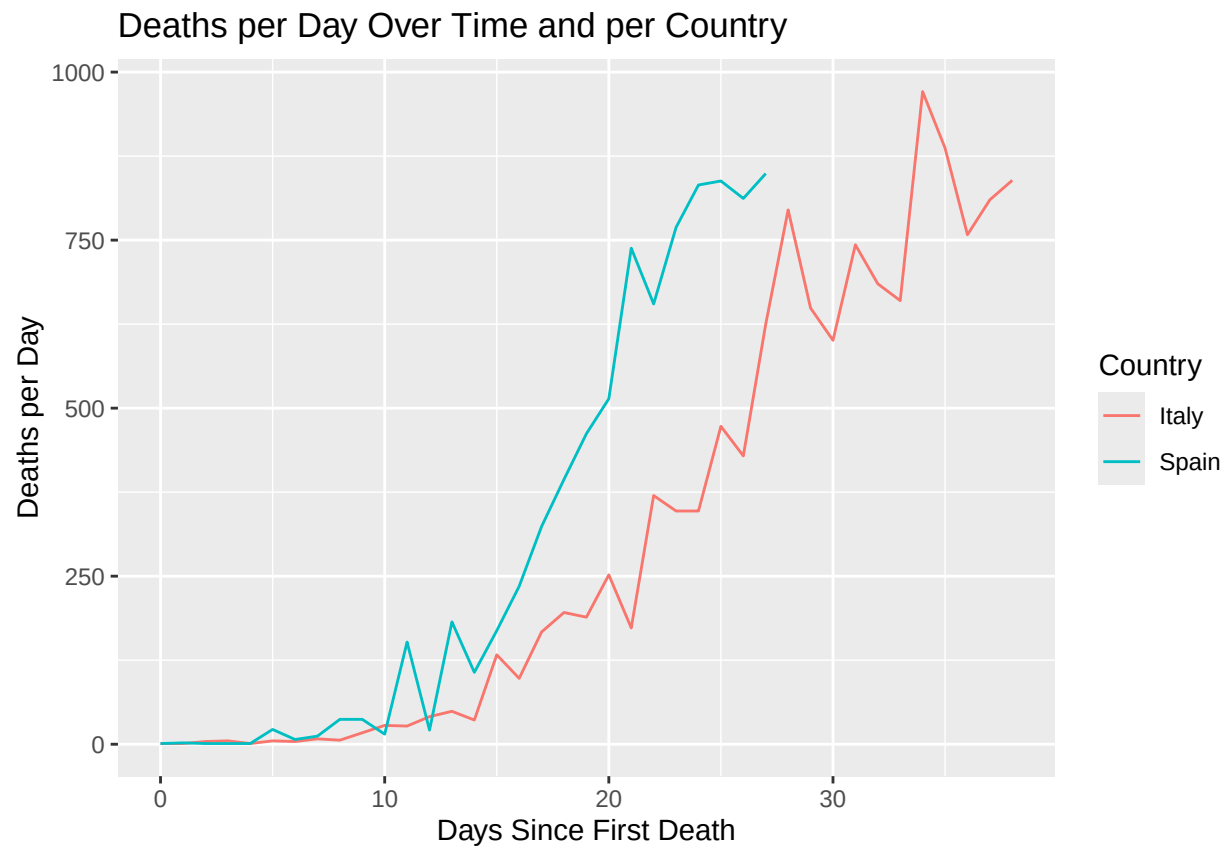
Spain's Growth Rate: 0.277

Spain's Growth Rate 95% Confidence Interval: [0.235, 0.318]

Spain's Doubling time: 2.5 days

Spain's Doubling time 95% Confidence Interval: [2.2, 3.0]

Part 3



A graph showing the deaths per country over time.

Problem 3

Economists use the power-law model $Q = KP^\beta$, where Q is quantity demanded, K is a constant, P is the price, and β is the price elasticity of demand, which is our desired result. By taking the log of both sides, we obtain the equation

$$\log(Q) = \log(KP^\beta) = \log(K) + \log(P^\beta) = \log(K) + \beta \log(P)$$

This mimics a linear regression model:

$$\log(Q) = \beta \log(P) + \log(K)$$

where β is the slope and $\log(K)$ is the constant intercept. By taking the log of price and sales and fitting a linear regression model, and then running a Monte Carlo simulation 10,000 times, we can obtain an estimate and confidence interval for the price elasticity of milk, which will be our slope.

Milk Price Elasticity Estimate (β): -1.617

95% Confidence Interval: [-1.775, -1.455]